

TREMORTRACE · DESIGN DECISION RECORD

TremorTrace TRL 2 review decisions

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/tremortrace		
Source	docs/decisions/0001-trl2-review-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-09-25	Amish Chadha	Record Amish's 2026-09-25 decisions on the TRL 2 review points

0001: TRL 2 review decisions

- **Date:** 2026-09-25
- **Status:** accepted (items D1 to D5); items O1 to O3 remain proposed

■ Context

The TRL 2 review note (docs/REVIEW.md , sessions of 2026-09-24) listed design choices marked "Proposed, awaiting Amish". On 2026-09-25 Amish reviewed the review points for every repo and wrote: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them." Every item that carried a recommendation is therefore decided as recommended. Items without a recommendation stay open.

■ Options considered

Table 1. Items with a recommendation.

#	Item	Options	Recommendation
D1	Controller	Single nRF52840 module with built-in IMU, flash and charger (Seeed XIAO nRF52840 Sense class); or a separate nRF52 module and IMU breakout	Single module, no custom PCB
D2	What the band stores	10 s summaries only; raw data only; summaries with optional raw snippets	Summaries, with optional raw snippets for research
D3	Phone route	Web Bluetooth page; native app	Web Bluetooth page first, native app later
D4	Which wrist	Most affected wrist; both wrists	Most affected wrist first, both later
D5	Data handling	Local-first, owner-controlled export; cloud service	Local-first, no cloud dependency (as R9 requires)

■ Decision

- **D1.** Decided by Amish, 2026-09-25: go with recommendation. One nRF52840 module with built-in LSM6DS3TR-C IMU, 2 MB flash and charger; no custom PCB for the first build.
- **D2.** Decided by Amish, 2026-09-25: go with recommendation. The band stores 32-byte 10 s summaries; a 256 kB area is reserved for raw snippets captured on demand (TRT-CAL-001, section 6).
- **D3.** Decided by Amish, 2026-09-25: go with recommendation. Sync and status use a web Bluetooth page first; a native app may follow.
- **D4.** Decided by Amish, 2026-09-25: go with recommendation. One band on the most affected wrist first; both wrists later.
- **D5.** Decided by Amish, 2026-09-25: go with recommendation. Data stays on the band and the owner's own devices; the owner controls any sharing.

Budget and pitch: the TRL 2 review made no recommendation to change either, so budget_usd stays at \$150 and the pitch and problem lines are unchanged.

Items that remain open (no recommendation was made, so they stay "Proposed, awaiting Amish"):

- **01.** First co-design and validation partner: a movement disorders clinic, a patient association, or the OpenRatio network. No recommendation; portfolio guidance is that community designs pick co-design partners per area later. Proposed, awaiting Amish.
- **02.** Confirmation of "OpenRatio" as a possible partner (part of O1). Proposed, awaiting Amish.
- **03.** Approval of kit 1.3.1, whose forearm-for-scale rule for small objects replaces the 1.75 m person in the concept media. No recommendation was stated in the review. Proposed, awaiting Amish.

New item raised at TRL 3 (not part of the 2026-09-25 decision):

- **04.** Charge current: the 50 mA setting (0.33 C, about 3.6 h, about 65 mW charger heat) or the 100 mA setting (0.67 C, about 1.8 h, about 130 mW). Recommendation: 50 mA, to keep the pod cool. Proposed, awaiting Amish.

■ Consequences

- TRT-REQ-001 v0.3: R5 now counts 7 days at 16 h of wear per day alongside a raw-snippet reserve; R11 names the LED and the web Bluetooth page; the assumptions name the most affected wrist.
- TRT-PRC-001 v0.4: D1, D2, D3 and D5 are recorded as decided; first-order numbers are replaced by TRT-CAL-001 values.
- TRT-PRB-001 v0.3: the wrist question is closed; the partner question stays open.
- The design stops at TRL 3. TRL 4 is on hold by Amish's instruction.